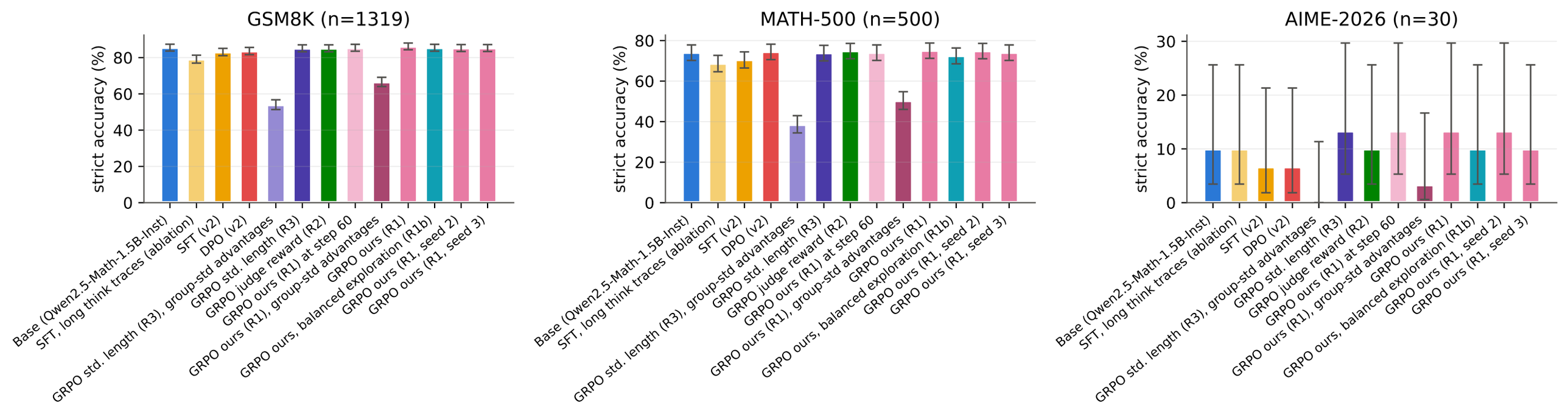
GSM8K (n=1319)
MATH-500 (n=500)
AIME-2026 (n=30)
strict accuracy (%)
Base (Qwen2.5-Math-1.5B-Inst)
SFT, long think traces (ablation)
SFT (v2)
DPO (v2)
GRPO std. length (R3), group-std advantages
GRPO std. length (R3)
GRPO judge reward (R2)
GRPO ours (R1) at step 60
GRPO ours (R1), group-std advantages
GRPO ours (R1)
GRPO ours, balanced exploration (R1b)
GRPO ours (R1, seed 2)
GRPO ours (R1, seed 3)